THE UNITED REPUBLIC OF TANZANIA

ORGANIZATION OF HEADS OF NON -GOVERNMENT SECONDARY SCHOOL

(OHONGSS)

KIGOMA REGION

FORM FOUR MOCK EXAMINATION

032/1 **CHEMISTRY 1**

TIME: 3Hours **YEAR: 2024**

1 This paper consists of section A, B and C with a total of **eleven** (**11**) questions.
2 Answer all questions in section A and B and two (2) questions from section C.
3 Section A carriers sixteen (16) marks, section B fifty four (54) marks and section C thirty (30) marks.
4 Non-programmable calculator may be used.
5 Cellular phones and any unauthorized materials are not allowed in the examination room.
6 Write your Examination Number on every page of your answer booklets.
7 The following constants May be used;
Atomic mass H=1, C=12, O=16, S=32, Zn=65 Ca=40, Na=23 Cu=64, Ag=108.

Gram molecular volume (G.M.V) at stp $22.4dm^3$

Avogadro's number $6.02X10^{23}$

IF=96500

IL=$1dm^3$=$1000cm^3$

SECTION A (16 Marks)

Answer **all** questions in this section.

1 For each of the items (i)-(x), choose the correct answer from the given alternatives and write its letter in the answer booklet provided.

i. Sodium hydroxide was added to three different test tubes X, Y and Z. Blue, green and reddish brown precipitates were formed respectively. The metal ions present in the test tubes X, Y and Z respectively were likely to be.....

A. Ca^{2+}, Pb^{2+} and Al^{3+}

B. Cu^{2+},Fe^{2+} and Fe^{3+}

C. Ag^{+}, $Pb^{2+}$and Mg^{2+}

D. Pb^{2+}, $Mg^{2+}$and Ca^{2+}

E. Zn^{2+}, Al^{3+} and Fe^{2+}

F.

ii. Which of the following factors affect the rate of heterogeneous reaction only?

A. Temperature

B. Nature of the reactants

C. Surface area of reactants

D. Concentration of reactants

E. Decreasing of concentration of reactants

iii. An element X is found in period 4 and group II of the periodic table. If element X undergoes the reaction $x \longrightarrow x^{2+}+2e$. The electronic configuration of X ion is.....

A. 2:8:6

B. 2:8:2

C. 2:8:8

D. 2:8:4

E. 2:8:8:2

iv. Which physical method can be used to separate the products of the following reaction?

$BaCl_{2\,(aq)}+Na_2CO_{3\,(aq)}) \longrightarrow BasO_{4(s)})+2NaCl_{\,(aq)}$

A. Evaporation

B. Filtration

C. Chromatography

D. Sublimation

E. Distillation

v. When chlorine gas is bubbled through distilled water; the distilled water turns pale yellow. Which of the following equations represents the reaction?

A. $Cl_{2g})+H_20_{(l)} \longrightarrow Cl_2.H_2O_{(aq)}$

B. $Cl_{2(g)} + H2O_{(l)} \rightleftharpoons HCl_{(aq)} +HOCl_{(aq)}$

C. $2Cl_{2(\,g)}(+2H_2O_{(l)} \rightleftharpoons 4Cl_{(aq)} +2H_2O_{(aq)}$

D. $Cl_{2(g)}+2H_2O_{(l)} \rightleftharpoons 2HOCl_{(aq)}+H_{2(g)}$

E. All of the above.

vi. John wants to electroplate a spoon with copper by using a copper sulphate solution. He should arrange the electrode in the following way;

A. Spoon as anode and copper as cathode
B. Spoon as cathode and copper as anode
C. Spoon as anode and carbon as cathode
D. Spoon as cathode and copper sulphate as anode
E. Spoon as cathode and carbon as anode

vii. Form Three students went to the chemistry laboratory and found a bottle containing a liquid. When that liquid was poured on the wool, it completely damaged the surface of the wool. Which among the following is best explanation of the meaning of a liquid found in the bottle?

A. It is any substance that contains oxygen in its formula
B. It is any substance that contains hydrogen in its formula
C. It is any substance that dissolves in air
D. It is any substance that dissolves in water to produce oxygen ion
E. It is any substance that exhibits different colors in different media

viii. Jitihada secondary School has three laboratories which are chemistry laboratory, physics laboratory and biology laboratory. In their chemistry laboratory you may find the following apparatuses EXCEPT....

A. Measuring cylinder, gas jar and retort stand.
B. Wire gauze, test tube rack and deflagrating spoon
C. Burette, pipette and conical flask
D. Microscope, drawing board and prism
E. Stopwatch, Thistle funnel and dropper

ix. One among the reasons that make luminous flame to be not suitable for heating purpose in the laboratory is that they produce soot. What is the reason for the production of the soot?

A. The carbon dioxide supply is usually very much enough
B. The flame produced is very hot

C. The carbon dioxide supply is usually not enough

D. The oxygen supply is usually much enough

E. The oxygen supply is usually not enough

x. Experts from Tanganyika Hospital were doing scientific research about cholera in Misungwi District council, and obtained the results from their experiment that do not support their hypothesis. What advice could you give to them?

A. The experiment should be changed

B. The result should be left out

C. Ideas for further testing to find a solution should be given

D. A new problem should be identified

E. The observation should be altered

2 Match the chemical equation in list **A** with the corresponding types of Chemical reactions in list **B** by writing the letter of the correct response beside the item number in the answer booklet provided.

LIST A	LIST B
i. $NH_{3(g)}+HCl_{(g)} \rightleftharpoons NH_4Cl_{(aq)}$	A. Endothermic reaction
ii. $C_{(s)}+O_{2(g)} \longrightarrow CO_{2(g)} + Heat$	B. Ionic reaction
iii. $2C_{(s)} +2H_{2(g)} +Heat \longrightarrow C_2H_{4(g)}$	C. Exothermic reaction
iv. $N_{2(g)} +3H_2(g) \longrightarrow 2NH_{3(g)}$	D. Neutralization reaction
v. $H_2SO_{4(aq)} +2KOH_{(aq)} \longrightarrow K_2SO_{4(s)} +2H_2O_{(l)}$	E. Reversible reaction
vi. $CaCO_{3\,(s)} \longrightarrow CaO_{(s)} +CO_{2\,(g).}$	F. Homogeneous reaction
	G. Displacement reaction
	H. Decomposition reaction.

SECTION B (54 Marks)

Answer **all** questions in the section

3 (a) People living near Geita gold mines faces the challenges on destruction of their infrastructures. As an expert assess three major ways which can be used to control environmental impacts that are caused by extraction of metal through mining.

(b) A teacher told form two students to write the balanced chemical equation for the action of heat on iron (ii) sulphate, action of heat on calcium hydrogen carbonate and action of heat on ammonium nitrate but they failed to do so because it is above their level. As a form four student help them to do so.

4 A form three Student wishes to measure the reaction rate of zinc with dilute hydrochloric acid. The chemical equation for the reaction is

$$Zn_{(s)}+2HCl_{(aq)} \longrightarrow ZnCl_{2(aq)}+H_{2(g)}.$$

In a given experiment, 0.016g of zinc granules was added into 1.00L of 0.100MHCl, and the following data were recorded.

Time(s)	0	4	8	12	16	20
Mass of zinc(g)	0.016	0.014	0.012	0.010	0.008	0.006

a. Calculate the rate of the reaction in grams of the zinc granules consumed per second.
b. Calculate the rate of the reaction in moles of the zinc granules consumed per second.
c. Write the complete ionic Equation for the reaction.
d. What would happen to the acid as the reaction proceeds?

5 (a) Two metals **A** and **B** are both divalent and have no effect on water**. A** is more electropositive than B. Explain in terms of electron transfer ,what you would expect to happen if **A** is placed in a solution of a salt of **B.**

(b) Explain why metals below hydrogen cannot react with acid to form hydrogen gas.

(c) P, Q, R, and S are metals. P reacts with steam whereas Q is not affected by either cold water or steam, R reacts violently with cold water while S bursts into flames as soon as it comes into contact with cold water. Arrange the metals in order of decreasing reactivity.

6 A colorless gas K fumes in the air. The gas gives dense white fumes when a glass rod dipped in ammonia solution is held near the glass.

(a) Name the gas K

(b) Name two chemicals used in the preparation of gas K

(c) Write a balanced chemical equation for the reaction of the chemicals named in 6(b)

(d) Why does the gas K fume strongly in air?

(e) Why does the gas K form dense white fume with ammonium hydroxide?

7 When a green substance **A** is heated, it gives off gas **B** which turns lime water milky and residue **C** which dissolve in the dilute hydrochloric acid to form a green Solution **D**. When a nail is dipped into solution **D,** a reddish brown coating **E** was deposited.

i. Identify and write the formula of substance **A, B, C, D** and **E**

ii. Write chemical equations for the formation of substance **B, C, D**, and **E**.

8 (a) Two organic compounds react together as summarized in the equation shown below.

$$C_2H_5OH_{(l)} + CH_3COOH_{(l)} \xrightarrow{Conc.H_2SO_4} CH_3COOC_2H_{5(l)} + H_2O_{(g)}$$

A **B** **C**

i. Name the compounds marked **A, B** and **C**

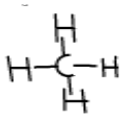
ii. What is the role of concentrated H_2SO_4 in reaction?

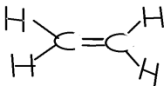
iii. Name the homologous series to which compounds **A** and **B** belong.

iv. State any two uses of compound **C**.

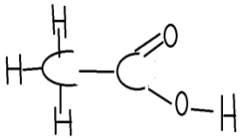
v. What type of Chemical reaction is portrayed by the equation?

(b)Study the following formulae of organic compounds then answer questions that follow.

A. 

B. 

C. CH3CH2OH

D. 

Given the name and write the homologous series to which each of the compounds belongs.

SECTION C (30 Marks)

Answer two (2) questions in this section

9 A doctor prescribed antacid tablets made from magnesium hydroxide to a student who was suffering from excess stomach acid.

a. Why did the doctor prescribe the antacid tablets?

b. Write a balanced chemical equation that took place in the student's stomach after taking the tablets?

c. If the normal acid content in the stomach is 160.00millimolar and the stomach secrets 0.50litres of gastric juice in a day, calculate the mass of the tablets to be given to the student if the gastric juice had 210.00millimolar hydrochloric acid.

d. If each tablet of an antacid weighs 145mg of Magnesium hydroxide, how many tablets will the student be given?

e. If a dose requires 1 tablet per every 12 hours, how many days should the student take to finish the full dose?

10 Farms at Lubuga village and Ndinga village lack some nutrients. Thus, villagers were advised to add inorganic fertilizers in their farms in order to obtain high yields. Villagers at Lubuga village accepted to add inorganic fertilizers in their farms but villager at Ndinga village were not willing to add inorganic fertilizers in their farms. Appraise six reasons to support villagers at Ndinga village.

11 (a) Eutrophication is excessive growth of aquatic plant and algae in water bodies;

i. Give two causes of eutrophication

ii. Explain the effect of eutrophication (give two points)

(b) Explain how industrial workers can be protected against harmful Effect of chemicals. Give three points.